Authored by : Arpit Raj , Lnmiit jaipur

Syntax

`MOD(dividend, divisor)`

Even/Odd Detection

The most common interview application.

Even numbers

```
SELECT *
FROM Students
WHERE MOD(student_id,2)=0;
```

Odd numbers

```
SELECT *
FROM Students
WHERE MOD(student_id,2)=1;
```

Difference between % and MOD()?

MOD() is a function and is more portable across SQL dialects, while % is an operator supported by some databases.

ABS() returns the absolute value of a number.

SQRT() computes the square root.

```
SELECT ROUND(SQRT(20),2);
```

↓

4.47

Very important.

```
SQRT(-25)
```

Mathematically impossible in real numbers.
Most databases return

NULL

Error

depending on the vendor.

RANDOM()

9.1 Introduction

Random number generation yntax

Database	Function
MySQL	`RAND()`
PostgreSQL	`RANDOM()`
SQLite	`RANDOM()`
SQL Server	`RAND()`

Random Integer

1–100

MySQL

```
SELECT FLOOR(RAND()*100)+1;
```

Random Row Selection

MySQL

```
SELECT *
FROM Students
ORDER BY RAND()
LIMIT 1;
```

ORDER BY RANDOM()

Easy but expensive.
The database assigns a random value to every row and sorts the entire result set.

Syntax

Natural logarithm

```
LN(number)
```

Base-10 logarithm

```
LOG10(number)
```

General logarithm (syntax varies by vendor)

```
LOG(base, number)
```

```
SELECT LOG(2,8);
```

↓

3